

- 3 A continuous random variable X has probability density function f given by

$$f(x) = \begin{cases} kx & 0 \leq x < 1, \\ k(8-x) & 1 \leq x \leq 8, \\ 0 & \text{otherwise,} \end{cases}$$

where k is a constant.

- (a) Show that $k = \frac{1}{25}$. [2]

[illegible]

- (b) Find the median value of X . [3]

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(c) Find the probability density function of Y .

[5]

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